

Causal inference on endogenous social network formation

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November 2025

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Introduction

- How do network ties form?
 - Based on exogenous factors (e.g. shared characteristics, place), observed and unobserved.
 - Based on existing ties (e.g. triadic closure).
- *Causal identification* for network formation is hard.
 - Unobservables, reverse causality, equilibrium.
- *Statistical inference* for networks is conceptually subtle.
 - Many small networks? Sampling from a large network?

- Why are there so many triangles?
 1. Homophily based on unobservables?
 2. Causal effect of indirect ties?
⇒ Which tie is the chicken, which the egg? (Reflection problem.)
- When is a network in equilibrium?
 1. No desired individual deviations? (Nash)
 2. No pairwise coordinated deviations? (Pairwise stability)
- What is the (population) estimand of interest?
 1. Hypothetical infinite super-network?
 2. Population of disconnected networks?

Literature

1. Mechanisms of tie formation

- *Triadic closure*: Simmel and Wolff (1964), Coleman (1988), Heider (2013).
- *Cumulative advantage*: Merton (1968), Barabasi and Albert (1999), DiPrete and Eirich (2006), DiMaggio and Garip (2012).
- *Local density*: Burt (2009), Uzzi (1997), Gargiulo and Benassi (2000).

2. Statistical network models: (Kolaczyk and Csárdi, 2014)

- Exponential random graph models.
- Stochastic block models.
- Dynamic models.

3. Structural econometric network models: Graham (2015), De Paula et al. (2018), Graham and De Paula (2020), Christakis et al. (2020).

4. Design-based inference: Neyman (1923), Abadie et al. (2020), Abadie et al. (2023).

Empirical application

- Network of employees at a global professional services firm.
- Edges $\approx$ employees working together.
- Employees choose their collaborators.
- *Random initial assignment:*
Within offices, new hires are randomly assigned to teams.
- *Network dynamics:*
We observe the evolution of the network over time.

Setup

Identification and inference

Empirical application

Discussion

Setup and notation

- Time periods $t = 1, 2$, individuals $i, j \in \{1 \dots n\}$.
- Adjacency matrices A^t with $A_{ij}^t \in \{0, 1\}$.
- Structural (causal) relationship $A^2 = f(A^1)$.
- Randomization of initial network: A^1 uniform from $\mathcal{A}$.
- Design-based identification and inference:
 - We condition on sample $\{1 \dots n\}$, and on potential outcomes f .
 - Only source of randomness: Sampling of A^1 from $\mathcal{A}$.

Assumption 1 (Setup)

- *Structural relationship*: $A^2 = f(A^1)$.
- *Panel data*: Both A^1 and A^2 are observed.
- *Randomization*: $P(A^1 = A|f) = \frac{1}{|\mathcal{A}|}$ for all $A \in \mathcal{A}$.
- *Exclusion restriction*: $d(A^1) = d \Rightarrow y(f(A^1)) = Y^d$ for all $A \in \mathcal{A}$.
- *Support*: $P(d(A^1) = d) > 0$.

Examples

- Outcome: $Y = A_{ij}^2$ - presence of a tie between i and j .
- Treatments: $D = d_{ij}(A^1)$.

- *Indirect tie:*

$$D = \mathbf{1} \left(\sum_k A_{ik}^1 A_{kj}^1 > 0 \right)$$

- *Network degree:*

$$D = \sum_k A_{ik}^1$$

- *Local network density:*

$$D = \frac{\sum_{k > k'} A_{ik}^1 A_{kk'}^1 A_{k'i}^1}{\sum_{k > k'} A_{ik}^1 A_{k'i}^1}$$

- Randomization:

$\mathcal{A}$ is the set of matrices obtained by swapping new hires within offices.

Assumption 2 (Randomization by permutation)

- *Permutations:*
 - For permutation π of $\{1, \dots, n\}$, A_π is the matrix with entries $A_{\pi(i), \pi(j)}$.
 - Let Π be an algebraic group of permutations. The set $\mathcal{A}$ is given by

$$\mathcal{A} = \{A_\pi : \pi \in \Pi\}.$$

- *Invariance and equivariance:*
 - Let $\mathcal{E}$ be a set of edges (i, j) ,
and let $d_{ij}(A)$ be characteristics of A for $(i, j) \in \mathcal{E}$.
 - For all $\pi \in \Pi$, $\mathcal{E}$ is invariant under π :
 $(i, j) \in \mathcal{E} \Rightarrow (\pi(i), \pi(j)) \in \mathcal{E}$.
 - For all $\pi \in \Pi$ and $(i, j) \in \mathcal{E}$, d_{ij} is equivariant under π :
 $d_{ij}(A_\pi) = d_{\pi(i), \pi(j)}(A)$.

Assumption 3 (Permutation of subset of nodes)

- Permutations are restricted to a subset $\mathcal{I}$:

$$\Pi = \{\pi \in \mathcal{S}_n : \pi(j) = j \ \forall j \notin \mathcal{I}\},$$

- where $\mathcal{S}_n$ is the *symmetric group* of all permutations of the set $1, \dots, n$.

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Lemma: Inverse probability weighting

- Denote $Y = y(A^2)$ and $D = d(A^1)$.

- Define the IPW estimator

$$\hat{Y}^d = Y \cdot \frac{\mathbf{1}(D = d)}{P(D = d)}.$$

- Under Assumption 1,

$$E \left[\hat{Y}^d | f \right] = Y^d.$$

Lemma: Equivariance of randomization probabilities

- Suppose that item 3 of Assumption 1 (randomization) and Assumption 2 hold.
- Denote $p_{ij}(d) = P(D_{ij} = d)$, and $P_{ij} = p_{ij}(D_{ij})$.
- Then for all $\pi \in \Pi$

$$p_{\pi(i)\pi(j)}(d) = p_{ij}(d), \quad \text{and} \quad P_{\pi(i)\pi(j)} \sim P_{ij}.$$

Proposition 1: Inverse probability weighted linear regression

- Denote $p_{ij}(d) = P(D_{ij} = d)$, and $P_{ij} = p_{ij}(D_{ij})$.
- Define

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \sum_{(i,j) \in \mathcal{E}} \frac{1}{P_{ij}} (Y_{ij} - D_{ij} \cdot \beta)^2,$$
$$\beta = \frac{1}{|\mathcal{E}|} \sum_{(i,j) \in \mathcal{E}} \left[\left(\sum_{d \in \mathcal{D}} d \cdot d' \right)^{-1} \cdot \left(\sum_{d \in \mathcal{D}} Y_{ij}^d \cdot d \right) \right].$$

- Under Assumptions 1 and 2,

$$E[\hat{\beta}|f] = \beta.$$

Sample average treatment effect

- Special case: Binary treatment.
- $D_{ij} = (1, X_{ij})$, with $X_{ij} \in \{0, 1\}$.
- Then

$$\beta_2 = \sum_{(i,j) \in \mathcal{E}} (Y_{ij}^1 - Y_{ij}^0)$$

is the sample average treatment effect.

- Support condition: For all $(i, j) \in \mathcal{E}$,

$$0 < P(X_{ij} = 1) < 1.$$

- Example: Triadic closure.

Proposition 2: Within regression

- Suppose $\mathcal{E} = \mathcal{I} \times \mathcal{J}$, where $\mathcal{I} \cap \mathcal{J} = \emptyset$.
- Consider the multisets $\mathcal{D}_j = [D_{ij} : i \in \mathcal{I}]$.
- Define

$$\begin{aligned}\hat{\gamma}_j &= \operatorname{argmin}_c \sum_{i \in \mathcal{I}} (Y_{ij} - D_{ij} \cdot c)^2, & \hat{\gamma} &= \frac{1}{|J|} \sum_{j \in \mathcal{J}} \hat{\gamma}_j, \\ \gamma_j &= \frac{1}{|I|} \sum_{i \in \mathcal{I}} \left[\left(\sum_{d \in \mathcal{D}_j} d \cdot d' \right)^{-1} \cdot \left(\sum_{d \in \mathcal{D}_j} Y_{ij}^d \cdot d \right) \right], & \gamma &= \frac{1}{|J|} \sum_{j \in \mathcal{J}} \gamma_j.\end{aligned}$$

- Under Assumptions 1, 2, and 3,

$$E[\hat{\gamma}|f] = \gamma.$$

Proposition 3: Randomization inference

- Consider the null hypothesis that Y_{ij}^d does not depend on d , for all $(i, j) \in \mathcal{E}$.
- For $\pi \in \Pi$, under the assumptions of Proposition 1, define the permuted estimator

$$\hat{\beta}_{\pi} = \operatorname{argmin}_{\beta} \sum_{(i,j) \in \mathcal{E}} \frac{1}{P_{\pi(i)\pi(j)}} (Y_{ij} - D_{\pi(i)\pi(j)} \cdot \beta)^2,$$

and the p-value

$$p^{\beta} = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \mathbf{1} \left(\hat{\beta} \leq \hat{\beta}_{\pi} \right).$$

- Under the null,

$$P(p^{\beta} \leq \alpha) \leq \alpha.$$

Proposition 3 continued

- Under the assumptions of Proposition 2, define similarly

$$\hat{\gamma}_{\pi} = \frac{1}{|J|} \sum_{j \in J} \operatorname{argmin}_c \sum_{i \in I} (Y_{ij} - D_{\pi(i)j} \cdot c)^2,$$

and the p-value $p^{\gamma} = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \mathbf{1}(\hat{\gamma} \leq \hat{\gamma}_{\pi})$.

- Under the null hypothesis, we again get $P(p^{\gamma} \leq \alpha) \leq \alpha$.

Computational implementation (1)

- Assume again: $\mathcal{E} = \mathcal{I} \times \mathcal{J}$, where $\mathcal{I} \cap \mathcal{J} = \emptyset$.
- Data can be stored in matrices / arrays of dimension $|I| \times |J|$:

$$Y = (A_{ij}^2)_{i \in \mathcal{I}, j \in \mathcal{J}}, \quad D = (D_{ij})_{i \in \mathcal{I}, j \in \mathcal{J}},$$

where $D_{ij} = d_{ij}(A^1)$.

- Assignment probabilities: Column means.

$$p_{ij}(d) = \frac{1}{|\mathcal{I}|} \sum_{i'} \mathbf{1}(D_{i'j} = d), \quad P_{ij} = p_{ij}(D_{ij}).$$

- Permutations π :
Column j remains constant, any permutation of the rows i is allowed.

Computational implementation (2): IPW

- “Instrument:” $Z_{ij} = \frac{1}{P_{ij}} D_{ij}$.
- Weighted regression:

$$C = \left(\sum_{i \in \mathcal{I}, j \in \mathcal{J}} Z_{ij} \cdot D'_{ij} \right)^{-1}, \quad B_{i,i'} = C \cdot \left(\sum_{j \in \mathcal{J}} Z_{ij} \cdot Y_{i'j} \right), \quad \hat{\beta} = \sum_{i \in \mathcal{I}} B_{i,i}.$$

- Randomization inference:

$$\hat{\beta}_{\pi} = \sum_{i \in \mathcal{I}} B_{\pi(i), i}.$$

Computational implementation (3): Within regression

- Define similarly:

$$C_j = \left(\sum_{d \in \mathcal{D}_j} d \cdot d' \right)^{-1}, \quad W_{ij} = C_j \cdot D_{ij}, \quad G_{i,i'} = \frac{1}{|\mathcal{J}|} \sum_j W_{ij} \cdot Y_{i'j}.$$

- Then

$$\hat{\gamma} = \sum_{i \in \mathcal{I}} G_{i,i}, \quad \text{and} \quad \hat{\gamma}_\pi = \sum_{i \in \mathcal{I}} G_{\pi(i),i}.$$

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Empirical setting

- Global firm in the professional services industry.
- Entry-level employees hired straight from degree programs.
- Work in project teams. Tie $\approx$ working together.
- Initial team assignment determined by an HR manager. Random within offices.
- Later team assignment based on an internal labor market. Junior employees aim to be recruited by senior colleagues.

Sample characteristics

For the set $\mathcal{E}$ of potential ties.

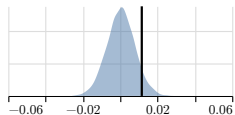
Variable	Mean	Std dev
Tie formed	0.0043	0.0653
N indirect ties	1.5726	4.5392
Degree	88.1196	62.4872
Local density	65.6269	19.3516
Indirect tie	0.3853	0.4867
High degree	0.5071	0.4999
High Density	0.5012	0.5000
Female	0.4613	0.4985
Black	0.0482	0.2143
Edu top 20	0.1131	0.3167

Placebo tests

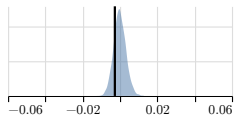
Outcome	New hires	Offices	Edges	Indirect tie	High degree	High density
Female	6,042	979	113,535,332	0.01155 (0.066)		
Black	6,042	979	113,535,332	-0.00276 (0.797)		
Edu top20	4,958	856	89,586,721	0.00576 (0.126)		
Female	4,414	490	105,968,417		0.03076 (0.046)	
Black	4,414	490	105,968,417		-0.00584 (0.774)	
Edu top20	3,589	421	87,098,920		-0.00647 (0.685)	
Female	5,161	654	118,682,052			-0.04057 (0.997)
Black	5,161	654	118,682,052			-0.00203 (0.618)
Edu top20	4,201	571	97,247,994			-0.00405 (0.676)

Permutation distribution: Placebo tests

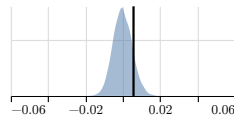
Indirect tie → female



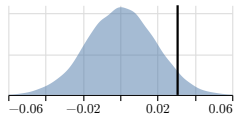
Indirect tie → black



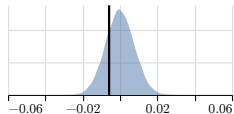
Indirect tie → edu top20



High degree → female



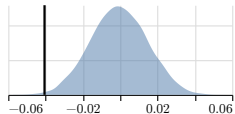
High degree → black



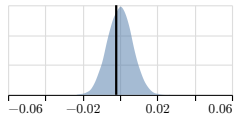
High degree → edu top20



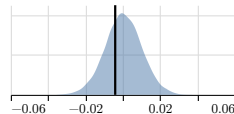
High density → female



High density → black



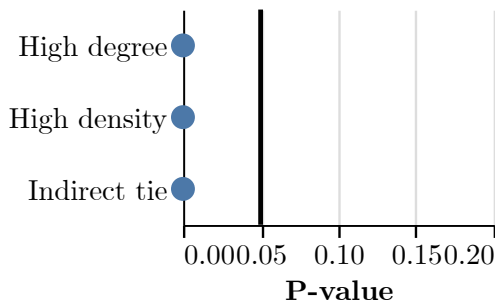
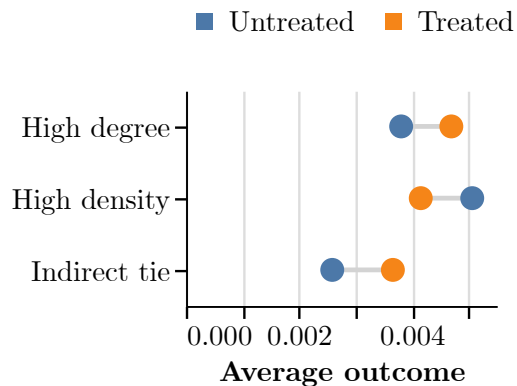
High density → edu top20



Estimates: Binary treatments

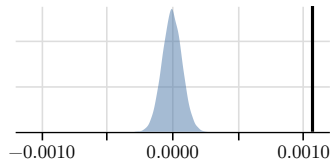
New hires	Offices	Edges	Intercept	Indirect tie	High degree	High density
6,042	979	130,686,467	0.0026	0.00107 (0.000)		
4,414	490	105,968,417	0.0038		0.00089 (0.000)	
5,161	654	118,682,052	0.0051			-0.00091 (1.000)
3,601	247	85,078,571	0.0023	0.00097 (0.000)	0.00020 (0.029)	-0.00031 (0.999)

Effect estimates

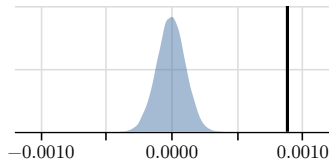


Permutation distribution: Effects

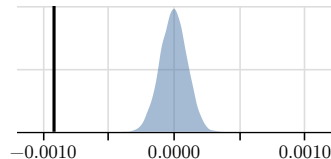
Indirect tie $\rightarrow$ tie formed



High degree $\rightarrow$ tie formed

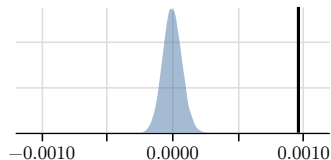


High density $\rightarrow$ tie formed

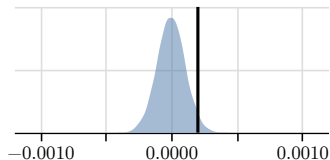


Indirect tie, High degree, High density $\rightarrow$ tie formed

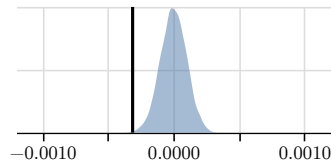
Indirect tie



High degree



High density



Estimates: Continuous treatments

New hires	Offices	Edges	Intercept	N indirect ties	Degree	Local density
6,042	979	127,738,642	0.0040	0.00056 (0.000)		
6,038	977	130,644,513	0.0034		0.00000 (0.386)	
6,036	976	130,665,151	0.0059			-0.00001 (0.562)
4,768	440	113,343,057	0.0065	0.00061 (0.056)	-0.00004 (0.791)	-0.00002 (0.617)

Setup

Identification and inference

Empirical application

Discussion

Discussion (1)

- All three of
 - indirect ties,
 - network degree,
 - local densityappear to matter for tie formation.
- The effect of indirect ties is quite robust and highly significant.
- The effects of network degree and local density are less robust.

Discussion (2)

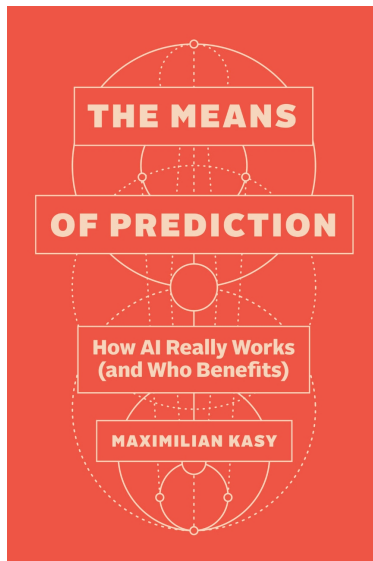
- *Design-based identification and inference:*
 - Avoids awkwardness of sampling models for networks:
Many small networks? Infinite super-network?
 - Avoids arbitrary asymptotics.
 - Based solely on partial randomness of initial ties.
- *Heterogeneity, super-populations, and estimands:*
 - No need to assume treatment effects are the same across ties.
 - No need to assume hypothetical super-population.
 - Inference for variants of the sample average treatment effect.

Discussion (3)

- *Dynamics versus equilibrium:*
 - Equilibrium notions for networks are ambiguous.
 - Who needs to consent to tie formation? Tie dissolution?
 - Additionally: Many equilibria - equilibrium selection?
 - Focusing on network dynamics allows us to avoid taking a stance.
- *Confounders and reverse causality:*
 - Confounders: Unobserved heterogeneity can easily lead to patterns like triadic closure, cumulative advantage.
 - Reverse causality: Did the tie between 1, 2 cause the tie between 2, 3, or the other way around? $\approx$ “reflection problem.”
 - Random initial assignment solves both.

On a separate note: My new book on the politics and economics of AI

https:
//press.uchicago.edu/
ucp/books/book/chicago/
M/bo255887145.html



Thank you!